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Prospective pilot evaluation of a deep learning model for kidney stone detection on CT using a web-based workflow platform

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Abstract

Rapid and reliable detection of kidney stones on non-contrast abdominal CT is essential for timely decision-making in emergency radiology. However, rising imaging volumes and workflow pressures continue to limit reporting capacity, creating a need for AI systems capable of supporting routine diagnostic practice. Although many AI-based stone detection models have been proposed, most rely on retrospective datasets, and few have been evaluated prospectively within environments that reflect real radiology workflow conditions. This study prospectively evaluates the performance, usability, and workflow compatibility of a deep learning–based kidney stone detection model deployed within a web-based platform designed to emulate key components of routine radiology practice, enabling forward-in-time evaluation without direct integration into routine clinical operations such as PACS/RIS or clinical reporting. A dual-stage convolutional neural network was developed using an internal dataset of 235 cases (3,452 slices) and validated through five-fold patient-level cross-validation. An independent set of 732 slices served as an independent hold-out set. For prospective evaluation, the trained model was integrated into a secure, browser-based interface supporting case upload, slice-level review, independent radiologist labeling, and visualization of AI-generated predictions. Over a six-month period, three radiologists uploaded and annotated a total of 5,152 anonymized CT slices. The platform dynamically calculated diagnostic metrics and logged human–AI interactions to assess performance stability and concordance. The pilot deployment demonstrated strong diagnostic performance under real-world variability, achieving 97.83% accuracy, 94.64% sensitivity, 98.27% specificity, 88.50% precision, and a Cohen's kappa of 0.90. Concordance between radiologists and the model exhibited increasing stability across sequential pilot stages. These findings present a reproducible framework for transitioning radiological AI systems from retrospective validation toward workflow-aligned, prospective pilot deployment. Although full PACS/RIS integration was not attempted, the results underscore the importance of pilot-stage evaluation as a critical intermediary step toward clinical implementation and regulatory approval.

Keywords Kidney stone detection · Urolithiasis · Non-contrast CT · Clinical workflow integration · Deep learning · Clinical decision support · Prospective pilot study

Introduction

The rapid increase in medical imaging demand, coupled with capacity constraints in radiology services, has considerably increased the workload of radiology departments in recent years. In the United Kingdom, for instance, approximately 976,000 imaging examinations in 2024 exceeded the National Health Service's one-month target and were not reported on time. These delays are largely attributed to the radiologist workforce remaining about 30% below the required level and an 8% annual increase in CT and MRI demand [1]. Similarly, the literature indicates that increasing imaging volumes and restricted reporting capacity prolong turnaround times, slowing diagnostic processes and limiting timely clinical decision-making [2–4].

Current capacity limitations and delays in radiology reporting have become important drivers motivating the transition of artificial intelligence (AI) applications from research environments to clinical practice. In the field of medical AI [5, 6], rapid advances in deep learning techniques have led to the emergence of numerous solutions for diagnostic tasks in radiology, resulting in a shift from

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research-focused use toward real-world clinical implementation. In this context, AI-based software solutions have begun to be gradually adopted in certain healthcare systems. For example, in the Netherlands, the number of radiology departments utilizing AI increased from 14 in 2020 to 23 in 2022, while the number of distinct products in use rose from 7 to 34, and total installations from 19 to 68 [7]. Moreover, it has been reported that the proportion of CE-marked AI products supported by evidence published in peer-reviewed journals increased from 36% in 2020 to 66% in 2023 [8]. While these advances indicate that AI technologies are gradually entering clinical environments, this transition is still nascent. Therefore, carefully designed pilot implementations are essential to systematically assess their performance, safety, and clinical impact before large-scale adoption.

Before adopting an AI solution on a large scale, conducting small-scale pilot studies within real clinical workflows is essential for understanding system performance, operational challenges, and user acceptance. Such pilots allow algorithms to be validated on independent patient cohorts, reveal unexpected sources of error, and provide opportunities for iterative refinement based on clinician feedback. Notably, several early pilot deployments have shown that AI systems can meaningfully influence radiology workflows, even if their benefits are often limited. For example, Sridharan et al. (2024) prospectively evaluated an AI-based chest X-ray triage tool embedded within an emergency radiology workflow and demonstrated that automated prioritization markedly reduced reporting turnaround time (TAT) by more than 70%, while maintaining acceptable specificity for urgent findings. Their results illustrated that AI can meaningfully reshape workflow efficiency even when the diagnostic component remains limited to triage rather than full report generation [9].

In contrast, Wenderott et al. (2024) examined an AI-assisted CAD system for prostate MRI in routine reporting and found that although clinicians perceived the tool as helpful and generally easy to use, the system did not significantly reduce reporting time or cognitive load, and its effect varied across PI-RADS categories. This underscores that AI tools integrated into more cognitively demanding diagnostic tasks may face challenges in generating measurable efficiency gains [10]. Similarly, Smith et al. (2025) implemented an AI-based stroke imaging support platform across a multicenter tele-stroke network and confirmed the feasibility and clinician acceptance of deploying multimodal CT decision support in urgent-care pathways. However, despite favorable usability feedback, key clinical metrics such as diagnostic consistency, decision-to-treatment time, and reperfusion rates did not show statistically significant improvement, highlighting the difficulty of influencing time-sensitive stroke workflows [11].

Other real-world studies have provided additional evidence on the opportunities and limitations of clinical AI deployment. Jones et al. (2021) integrated a multi-finding chest radiograph AI system into routine radiology reporting workflows over a six-week period and observed high agreement between AI outputs and radiologist interpretations (≈86%). The system prompted changes in approximately 3% of reports and modest modifications in patient management, demonstrating that AI can influence clinical decisions even when the overall magnitude of impact is limited [12]. van Beek et al. (2022) evaluated an ML-based CXR tool across both emergency and primary-care populations in a large, consecutive real-world cohort and reported consistently high performance across different clinical settings. Although workflow efficiency outcomes were not formally assessed, the study provided strong independent validation evidence supporting model robustness [13]. Meanwhile, Miró Catalina et al. (2024) assessed an AI algorithm for thoracic pathology screening in a primary-care radiography environment and demonstrated high specificity but notably limited sensitivity across several findings. Their results emphasized the need for further model refinement before routine clinical use can be considered [14].

Despite these encouraging pilot outcomes and growing evidence of feasibility, studies that have achieved full workflow integration of artificial intelligence (AI) tools remain scarce. This scarcity indicates that the transition from experimental prototypes to sustainable clinical adoption remains at an early stage. Accordingly, the full incorporation of AI solutions into routine clinical practice continues to face substantial challenges. The literature highlights several key barriers that hinder the successful implementation of AI systems in clinical environments, including the following:

- *Lack of trust and insufficient awareness or training regarding AI among radiologists and clinicians*: Some physicians remain skeptical about the diagnostic performance of AI tools and even express concern that such technologies may diminish their professional roles. This lack of confidence and difficulty in adoption significantly slows the acceptance and integration of AI applications into clinical practice [15, 16]. For example, some studies [12, 14] reported notably reduced sensitivity for specific findings or minimal influence on final clinical decisions, which can further undermine clinician trust and reinforce skepticism toward routine AI usage.
- *Limited evidence demonstrating the impact of AI systems on patient outcomes*: The lack of robust evidence showing that AI solutions can improve clinical outcomes reduces both investment and user willingness to adopt these technologies. Some users remain hesitant to implement AI in routine practice, believing that its practical contribution to clinical decision-making

may be limited [15, 16]. In this regard, multiple pilot deployments [10–12] reported no statistically significant improvement in reporting time, diagnostic consistency, or treatment-related metrics, illustrating the persistent uncertainty regarding AI's measurable clinical benefit.

- AI tools frequently operate as standalone applications rather than being fully integrated into picture archiving and communication systems (PACS) or radiology information systems (RIS). This disconnection disrupts clinical workflows, increases cognitive load for radiologists, and limits routine usage in time-sensitive diagnostic settings [17].
- The development and validation of reliable AI models depend on large, high-quality, and accurately labeled datasets. However, clinical datasets may suffer from incomplete annotations, inconsistent labeling standards, and selection bias, reducing the reliability and reproducibility of trained models [18].
- Regulatory and ethical concerns—including accountability for AI-generated decisions, uncertainty regarding legal liability in the event of diagnostic errors, and issues related to patient data privacy—represent additional major barriers. The lack of a clearly defined legal framework and the complexity of approval and certification processes further slow the integration of AI tools into hospital environments [15, 16].

Therefore, overcoming these barriers requires not only technical improvements but also the development of workflow-aligned systems that integrate seamlessly into PACS/RIS environments, thereby reducing the burden of standalone applications and minimizing workflow fragmentation. Strengthening communication and collaboration between clinicians and AI developers, establishing continuous training programs, and adopting model-development strategies that emphasize transparency, explainability, and robust performance across diverse anatomical structures are equally important for improving clinician trust. Addressing limitations related to data quality—such as incomplete annotations, inconsistent labeling, and dataset imbalance—will be essential for enhancing model reliability and mitigating performance variability observed in prior studies. Furthermore, generating strong, reproducible, and clinically meaningful evidence through multi-center and longitudinal evaluations is critical for demonstrating real-world clinical impact, given that many existing pilot studies have shown limited influence on reporting time, diagnostic consistency, or patient management. In parallel, clarifying regulatory frameworks and establishing clear ethical and legal standards regarding accountability, liability, and data privacy will support the safe and effective deployment of AI systems in clinical workflows. When these conditions are met, the potential benefits of AI technologies can be more reliably and sustainably translated into real clinical environments.

In this context, an essential intermediate step toward sustainable clinical adoption involves the creation of workflow-aligned, user-centric platforms that allow AI models to be prospectively evaluated within realistic, simulation-based diagnostic environments. Unlike many prior studies in which AI systems were deployed directly into clinical workflows, the present work introduces a structured transitional stage: a secure, web-based interface designed to emulate key components of radiologist workflow while enabling controlled, iterative assessment. This approach bridges the gap between model development and routine clinical deployment by allowing radiologists to interact with the system in real time, submit annotations, and immediately reviewing model outputs. Importantly, the implementation supports a bias-reducing sequence in which the AI prediction is revealed only after the radiologist's initial interpretation, ensuring independent assessment. In parallel, continuously updated performance metrics—refreshed with each newly annotated case—provide transparent, cumulative feedback that enables both developers and clinicians to monitor reliability, identify failure modes, and progressively build trust in the system.

As illustrated in Fig. 1, the present study follows this translational pathway by first developing an AI-based kidney stone detection model over a 12 month period using internally collected CT data and expert-verified annotations and subsequently embedding the model into the simulation-based interface for prospective testing. The six-month pilot deployment enabled longitudinal evaluation under realistic diagnostic conditions without the risks associated with immediate hospital-wide integration. This setup allowed the assessment of usability, workflow compatibility, and reliability in a controlled yet clinically meaningful environment, thereby offering critical insight into how the system may behave once fully implemented within a PACS-based infrastructure.

Kidney stone detection was selected as a clinically meaningful test case due to its high prevalence, the substantial review burden associated with non-contrast abdominal CT, and the need for rapid, observer-independent interpretation in emergency settings. Although identifying stones on CT is generally straightforward, radiologists often review hundreds of slices per patient, increasing cognitive fatigue and extending reporting time. Missed or delayed detections can further lead to unnecessary repeat imaging, prolonged patient discomfort, and delayed therapeutic decision-making. Consequently, AI systems capable of accurately localizing and flagging stone-positive slices hold considerable potential to reduce radiologist workload, accelerate diagnosis, and enhance the overall efficiency of urological imaging workflows. To contextualize this clinical demand and the transformative role AI approaches may serve in this domain,

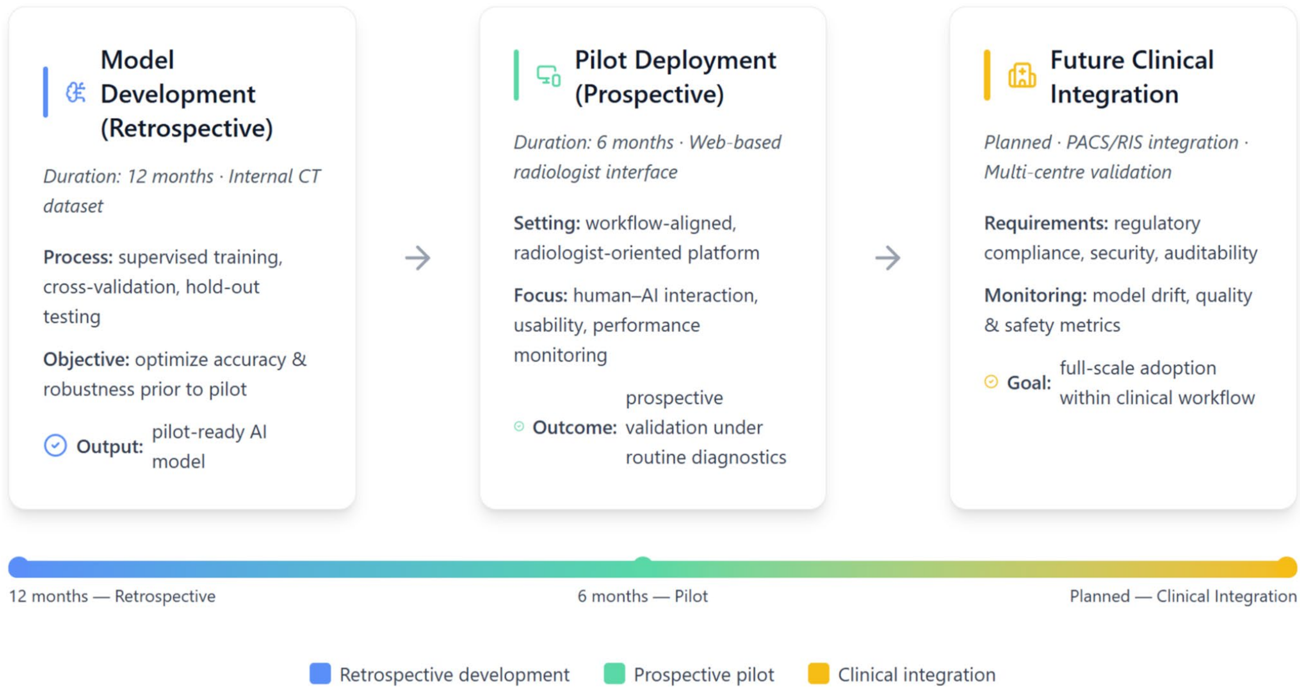

Fig. 1 Conceptual framework illustrating the translational pathway of the proposed AI-assisted kidney stone detection system, progressing from retrospective model development to a prospective web-based pilot and toward future clinical integration

the following subsection (Sect. "Introduction") provides a detailed discussion of the necessity and anticipated impact of AI-based kidney stone detection.

AI necessity in kidney stone detection

Urolithiasis is a common clinical condition characterized by a high recurrence risk and a substantial burden on healthcare systems. Epidemiological studies have reported that the global prevalence of kidney stones ranges between 1 and 20% [19], while in Turkey, this rate has been reported to be approximately 11.1% [20]. It is estimated that around two million individuals visit emergency departments each year due to severe pain caused by kidney stones (renal colic) [21]. In recent years, a notable increase in the number of patients presenting with suspected urolithiasis has been accompanied by a parallel rise in CT examinations, thereby increasing the workload in radiology departments and elevating the risk of diagnostic delays in cases requiring urgent intervention.

In the evaluation of patients presenting with suspected renal colic, non-contrast computed tomography (NCCT) is frequently preferred as the first-line imaging modality due to its high sensitivity and specificity [22, 23]. Non-contrast abdominal CT is considered the gold standard for the diagnosis of kidney stones and demonstrates exceptionally high performance in detecting urolithiasis, including small calculi. However, the detailed examination of numerous CT slices and the manual measurement of stone dimensions constitute a time-consuming and repetitive task for radiologists, potentially increasing workload and prolonging reporting times in busy clinical settings. Yet, early and accurate diagnosis is crucial for determining appropriate treatment strategies and preventing serious complications such as obstruction, renal damage, or infection [23, 24].

In line with these needs, artificial intelligence (AI)-based systems for kidney stone detection have emerged as potential solutions to support clinical workflows. Deep learning–based AI models can automatically identify and localize kidney stones on non-contrast CT slices and perform volumetric measurements. By eliminating the need for manual measurements, these systems reduce radiologists' workload, accelerate the evaluation process, and enhance clinical efficiency. Indeed, previous studies have demonstrated that applying AI methods for kidney stone detection on abdominal CT can improve routine diagnostic workflows while maintaining high diagnostic accuracy and providing substantial support to radiologists. For instance, a recent study reported that a deep learning–based system was capable of automatically detecting kidney stones and performing volumetric assessments on non-contrast CT, thereby automating repetitive tasks and improving clinical productivity [25].

In conclusion, AI-assisted systems for kidney stone detection hold significant potential to accelerate the diagnostic process, reduce radiologists' workload, and enhance the quality of patient care in high-demand clinical environments. Accordingly, the development of AI-based approaches for automated kidney stone detection, followed by validation through pilot clinical evaluations, represents a natural component of the ongoing digital transformation in radiology.

Contribution

The present study offers several notable contributions that extend beyond conventional retrospective AI evaluations and advance the translation of kidney stone detection models toward real-world clinical use:

- The study introduces a simulation-based transitional stage situated between retrospective model development and full clinical deployment. By enabling prospective evaluation in a workflow-mimicking environment, this approach addresses a major gap in the literature, where most prior studies integrate AI directly into clinical workflows without an intermediate validation layer.
- A prospective six-month pilot study involving 5,152 anonymized non-contrast CT slices was conducted within a workflow-simulated web interface, enabling realistic human–AI interaction, slice-level annotation, and continuous performance monitoring under near-operational conditions.
- The AI system was deployed within a secure, browser-based clinical interface that allowed radiologists to upload, review, annotate, and visualize AI predictions directly through the platform. The interface operated in parallel with standard reading practices, demonstrating workflow compatibility without requiring infrastructural or procedural modifications.
- The AI model—originally developed and internally validated over a 12-month period on a curated institutional dataset—was subsequently embedded into a modular, containerized Streamlit-based web platform. This architecture ensured portability, scalability, and compliance with hospital-grade IT requirements, enabling secure, real-time inference during the pilot phase and providing a deployment-ready foundation for future PACS/RIS integration.
- Diagnostic performance was evaluated across six sequential pilot assessment steps, demonstrating stable and consistently high AI–single reader agreement over time (κ from 0.75 to 0.90), supporting temporal robustness of the deployed system under prospective, workflow-simulated evaluation without retraining.

To contextualize these contributions within the broader landscape of real-world AI deployments in medical imaging, Table 1 summarizes representative prospective pilot studies reported in the literature and positions the present work among existing efforts. The table outlines how previous implementations have differed in terms of clinical setting, workflow integration, evaluation length, task complexity, and measurable impact, thereby providing a clear comparative backdrop against which the methodological scope and design choices of the present study can be interpreted.

The remainder of this paper is organized as follows. Sect. "Materials & Methods" describes the dataset, system architecture, and implementation details of the workflow-integrated AI platform. Sect. "Experiments" presents the experimental setup and performance evaluation procedures. Sect. "Discussion" discusses experimental results and key findings. Finally, Sect. "Conclusion" concludes the study and highlights directions for future research.

Materials & methods

Study design and workflow

This study was designed as a two-phase prospective pilot integration framework. Here, *prospective* denotes forward-in-time data collection within a simulation-based radiology workflow rather than embedded routine clinical operations. The study aims to evaluate the clinical usability and workflow compatibility of an AI-based system developed for automatic kidney stone detection on non-contrast abdominal CT slices. The first 12 month phase focused on development, offline validation, and optimization of the AI model, whereas the subsequent 6-month period involved deployment of the trained model into a web-based clinical interface and pilot testing with radiologists in a simulated real-world environment.

During the pilot deployment phase, the AI model was integrated into a secure web-based platform that enabled radiologists to review CT cases, manually label kidney stone presence, and subsequently view the system's automated predictions. The platform collected radiologist inputs, displayed model outputs, and dynamically computed performance metrics, enabling continuous monitoring of the model's generalization ability under routine-like diagnostic conditions. The workflow was designed to reflect key steps in standard radiology practice—case access, slice inspection, and diagnostic decision-making—except for formal report documentation, which was intentionally omitted. This was because the primary objective of the pilot implementation was not clinical reporting but evaluating the model's behavior in real-use scenarios, observing radiologist–AI concordance, and building user confidence through transparent performance visualization. Radiologists were encouraged to annotate cases independently; only after submitting their labels were the model's predictions revealed. This human–AI interaction strategy enabled systematic assessment of diagnostic agreement, user behavior, and potential workflow benefits. All interactions—including radiologist

Table 1 Representative prospective pilot studies of imaging-based AI systems in clinical or workflow-related settings, along with the present study

Study	Imaging modality & target task	Study design & clinical setting	Workflow /deployment focus	Key finding	Pilot duration
Jones et al. (2021) [12]	Chest X-Ray/Multi-finding DL-based diagnostic support	Prospective multicenter real-world deployment in routine chest radiography workflow	DL diagnostic support tool integrated into daily clinician workflow/Diagnostic accuracy & report times evaluated	Demonstrated high diagnostic agreement between AI and radiologists (≈86%), resulted in measurable changes in clinical reports and patient management, and was perceived by clinicians as useful and acceptable for CXR interpretation	~6 weeks
Van Beek et al. (2022) [13]	Chest X-Ray/Pneumonia and major thoracic abnormality detection	Consecutive real-world cohort (ED + primary care)	ML-based CXR software applied to consecutive patient images across settings/Clinical validation across sites	High and consistent performance demonstrated across settings; validated in real patient cohort	Not specified
Wenderott et al. (2024) [10]	Prostate MRI/AI-based CAD for lesion detection	Prospective pre–post evaluation embedded in routine prostate MRI reporting workflow	Integration of AI-based CAD into standard radiology workflow/ Reporting time and workload effect examined	No significant reduction in reporting time or radiologist workload; CAD did not improve workflow efficiency	3 months
Sridharan et al. (2024) [9]	Chest X-Ray/Triage (normal/non-urgent/urgent)	Prospective real-world clinical study	AI-based CXR triage system embedded in emergency and general radiology workflow	Reporting turnaround times reduced by > 70%; high specificity and useful in real-world setting	4 weeks
Miró Catalina et al. (2024) [14]	Chest X-Ray/ Primary care screening of broad thoracic pathologies	Real-world testing in primary care health centers	AI algorithm applied in primary care radiography workflow for general thoracic pathology detection	AI demonstrated high specificity but limited sensitivity; clinical usability remains uncertain and requires further improvement	~6 weeks
Smith et al. (2025) [11]	CT/perfusion + vascular imaging/Acute ischemic stroke imaging support	Prospective evaluation/Multi-center tele-stroke network	Deployment of imaging AI support in tele-stroke workflow/Decision times and patient flow assessed	Demonstrated feasibility of AI-driven support in a multi-center tele-stroke network; however, no statistically significant improvement in workflow metrics or treatment times was observed	3 months pre-implementation + 3 months post-implementation
Present work	Non-contrast CT/ Kidney stone detection	Prospective evaluation/Web based simulated radiology workflow	AI model integrated into modular browser-based platform/Radiologist annotation, review, and real-time inference supported	Improved model–radiologist agreement (κ = 0.75 → 0.90); reduced manual slice-review burden via automated kidney region filtering; demonstrated compatibility within a simulated radiology workflow	6 months

labels, model predictions, and time stamps—were logged by the platform, providing a comprehensive framework for evaluating system performance throughout the six-month pilot period.

Overall, this pilot deployment framework served to bridge controlled offline evaluation with real-time user interaction, with the goal of identifying practical considerations, usability challenges, and integration requirements prior to large-scale clinical application.

System architecture

The proposed system was implemented as a modular web-based platform designed to embed the trained AI model into a radiologist-oriented diagnostic workflow. The architecture follows a layered structure—comprising Access, Data, Presentation, Service, and Model layers—that enables secure user authentication, case management, slice visualization, AI inference, and performance monitoring, as illustrated in Fig. 2.

Access layer

The access layer provides secure credential-based login for radiologists and ensures controlled user authorization. Role-based access management was implemented to maintain data integrity and prevent unauthorized modifications. User identity and session information were managed through encrypted tokens.

Data layer

This layer handles CT study loading, case assignment, and storage of radiologist annotations. Uploaded radiology cases were displayed to the user through the interface, and corresponding labels were saved in a structured database. Annotation updates were version-controlled to enable longitudinal performance tracking and retrospective evaluation.

Presentation layer

The presentation layer serves as the user interface where radiologists review CT slices, annotate stone presence, view AI predictions, and monitor performance metrics. The interface provides slice-by-slice visualization and annotation controls. A performance dashboard displays key evaluation metrics—including accuracy, sensitivity, specificity, F1-score, and Cohen's kappa—updated in real time as additional cases are annotated.

Service layer

The service layer orchestrates the interaction between the radiologist interface and the embedded AI model. Upon annotation submission, the system sends the CT slice to the inference engine, retrieves the model prediction, compares radiologist and model outputs, and updates performance statistics. Logging and audit mechanisms capture case identifiers, timestamps, annotation details, and disagreement records for later analysis.

Model layer

The AI model, previously trained and validated offline, follows a dual-stage, cascaded multi-task convolutional architecture that separates anatomical localization from slice-level classification. In the first stage, an image-to-region localization module, inspired by DeepLabV3 + [26], performs image-to-image regression to directly transform the full non-contrast CT slice into a ROI-preserved representation, in which renal regions are emphasized while non-renal areas are suppressed. Importantly, no explicit segmentation mask, bounding box, or cropping operation is produced or applied. Instead, anatomical localization is achieved implicitly through regression-based image transformation while preserving the original spatial context. The resulting ROI-preserved slice is then forwarded to the classification stage without tight ROI extraction.

Both the localization and classification networks were designed to accept 224×224 input slices. The classification component adopts a dual-encoder architecture, in which a deep encoder with frozen ImageNet-pretrained weights captures high-level semantic representations, while a shallow encoder, whose depth was determined via hyperparameter optimization, focuses on fine-grained texture and edge-level features. The fused representations enable robust discrimination under heterogeneous anatomical and imaging conditions. Detailed architectural specifications of the classification subnetwork are reported in our prior work [26], whereas the localization module was developed internally to support the cascaded design used in this pilot study.

The model was trained using the Adam optimizer with a learning rate of 1×10^{-4} for a maximum of 50 epochs, with early stopping applied based on validation loss. Online data augmentation, including random flipping, small-angle rotation, and scaling, was employed during training to improve generalization. A fixed decision threshold, determined during the development phase, was used for inference and remained unchanged throughout the prospective pilot deployment.

During operation, the model generated binary predictions accompanied by confidence scores. The complete pipeline

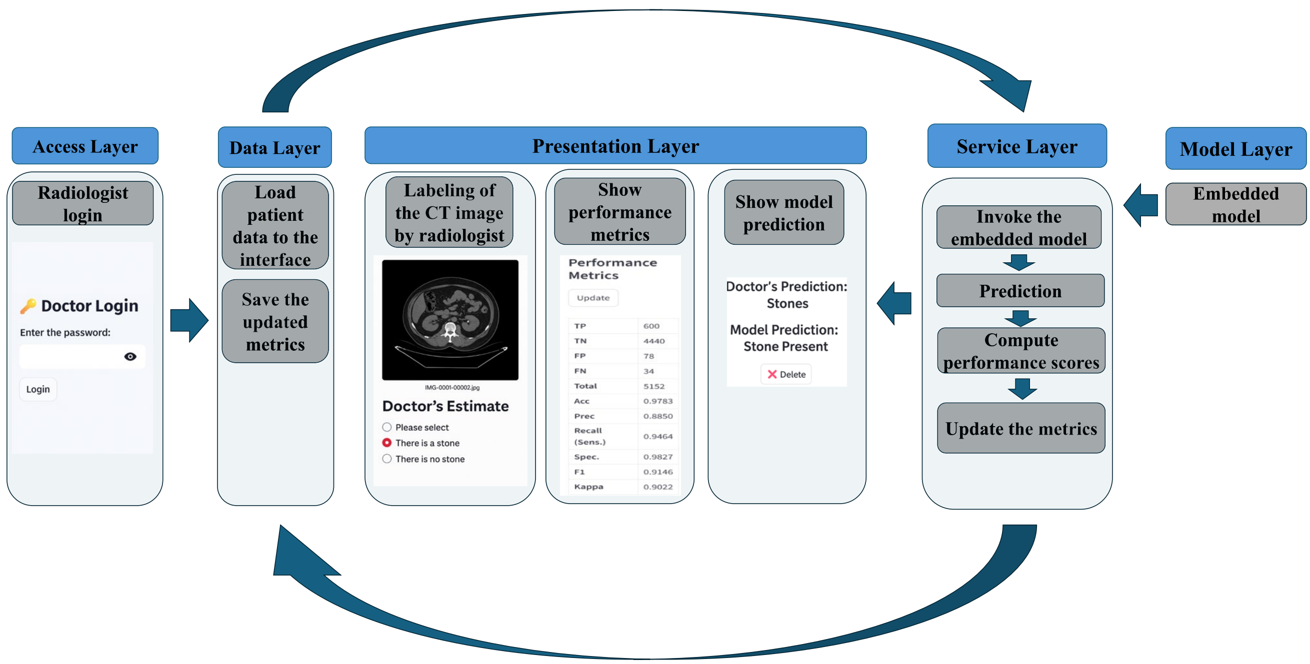

Fig. 2 Layered system architecture of the proposed AI-assisted radiological decision support framework

was embedded into the server-side execution layer of the pilot platform for real-time inference. To emulate standard hospital infrastructure, the inference service was containerized and optimized for CPU-based deployment, without reliance on GPU resources.

Procedural flow of the layered architecture in the pilot platform

This subsection describes the inter-layer operational workflow of the web-based pilot platform, detailing how the five architectural layers interact during case processing and performance evaluation. In Fig. 3, the interlayer flow of the proposed interface is explained through a pseudocode representation, illustrating the procedural interaction among the five architectural layers—Access, Data, Presentation, Service, and Model—defined in Fig. 2. The Access Layer manages radiologist authentication and secure system entry. Once logged in, the Data Layer handles the loading of anonymized CT studies (study) and updates cumulative performance metrics (C), as can be inferred from the procedural flow illustrated in Fig. 3. The Presentation Layer governs all user interactions, including slice-level review and labeling (y_r) performed by the radiologist (r), as well as the visualization of AI predictions (y_m) and metric dashboards. Within the Service Layer, the embedded AI model (M) is invoked for each CT slice (s) to generate predictions, after which the confusion matrix components are updated to compute diagnostic metrics. Finally, the Model Layer encapsulates the pretrained convolutional neural network responsible for stone detection and operates as the computational engine called by the Service Layer. Throughout this procedure, r denotes the participating radiologist, s the current CT slice under review, y_r and y_m the respective labels produced by the radiologist and the AI model, and C the cumulative set of performance counters dynamically updated after each inference. This structure ensures a clear mapping between human input, automated inference, and system-level feedback, enabling transparent and traceable evaluation during the pilot study.

Figure 4 presents the user interface of the web-based pilot platform, illustrating the key components through which radiologists interacted with the system during the study. As shown in Fig. 4(a), the access layer provided a secure login screen through which each radiologist authenticated before initiating case reviews. After successful login, users were directed to the main interface shown in Fig. 4(b), where anonymized non-contrast CT examinations could be uploaded via the file upload module. The same panel also displayed real-time performance metrics—including accuracy, sensitivity, specificity, precision, F1-score, and Cohen's κ—which were automatically updated as additional cases were processed.

For each uploaded case, radiologists sequentially reviewed the CT slices and assigned a binary label (stone present/absent) based solely on their independent visual assessment. Once the annotation was submitted, the system revealed the model's prediction for the same case within the interface, enabling immediate comparison while preventing any influence on the radiologist's initial decision. This *human-first* interaction design maintained diagnostic independence while allowing prospective monitoring of clinician–AI agreement under workflow-like operating conditions.

Dataset and annotation protocol

The AI model evaluated in this study was previously developed using a curated internal dataset of non-contrast abdominal CT slices collected during the first two months of the project. This dataset was constructed according to predefined inclusion criteria (patients aged 18–65 undergoing non-contrast abdominal CT for any reason) and exclusion criteria (patients over 65, cases with severe imaging artifacts, and conditions such as double J catheter, percutaneous nephrostomy, single kidney, congenital anomalies, atrophic kidney, calcified cysts, regional lymph node calcification, or medullary nephrocalcinosis). In the development phase, the model was trained and evaluated using five-fold cross-validation on a total of 3,452 CT slices obtained from 235 cases, consisting of 1,726 stone-positive slices from 196 patients and 1,726 stone-negative slices from 39 healthy subjects. Here, the term "healthy" denotes stone-free patients who underwent abdominal CT for clinical indications, rather than asymptomatic screening controls. The folds were constructed using a strict patient-level separation to ensure that slices from the same individual did not appear in both training and validation sets during cross-validation. An independent hold-out set of 732 slices (440 stone-positive slices from 45 patients and 292 stone-negative slices from 7 healthy subjects) was used for final hold-out evaluation. The independent hold-out set was drawn from the same institution as the development data but consisted of a separate patient cohort that was not used during training or cross-validation.

CT data were retrospectively collected from patients who underwent non-contrast abdominal CT examinations performed under a standard stone imaging protocol in a routine clinical setting. All scans were acquired using a commercial multi-detector CT system (GE Optima CT 660). Imaging was performed with a tube voltage of 120 kVp and automatic tube current modulation, typically ranging between 100 and 200 mA. Axial slices were reconstructed with a slice thickness of 2.5 mm. Information regarding the reconstruction

Fig. 3 Conceptual pseudocode summarizing the operational workflow of the radiologist–AI interface as implemented in the pilot study

Algorithm 1: Layered workflow of the radiologist–AI interface

```
Input: Embedded model M, initial performance counters C
Output: Updated performance metrics C
while pilot period is active do
    // Access Layer: radiologist login
    r ← RadiologistLogin() ;
    Authenticate(r) ;
    // Data Layer: load patient data to the interface
    case ← LoadPatientData(r) ;
    slices ← GetSlices(case) ;
    // Presentation Layer: labeling of the CT image by
       radiologist
    foreach slice s in slices do
        y_r ← LabelCTImage(r, s) ;
        // Service Layer: invoke embedded model and obtain
           prediction
        y_m ← InvokeEmbeddedModel(M, s) ;
        // Service Layer: compute performance scores
        UpdateConfusionCounters(C, y_r, y_m) ;
        // Presentation Layer: show model prediction
        ShowModelPrediction(s,y_r, y_m)
    // Service Layer: update aggregated metrics
    RecomputePerformanceMetrics(C) ;
    // Presentation Layer: show performance metrics
    ShowPerformanceMetrics(C) ;
    // Data Layer: save the updated metrics
    SaveUpdatedMetrics(C) ;
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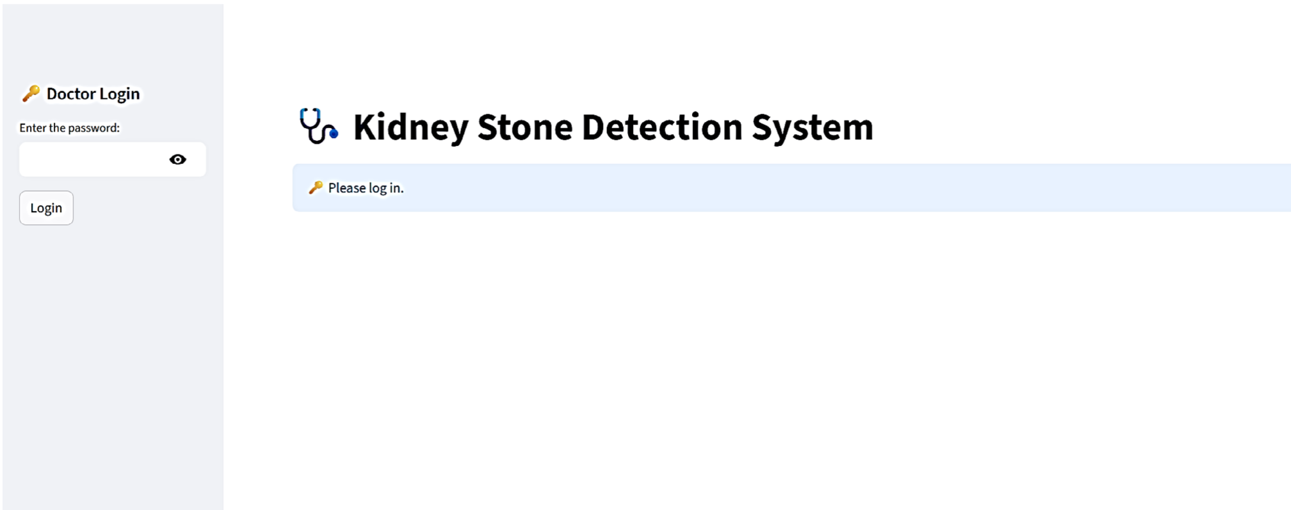
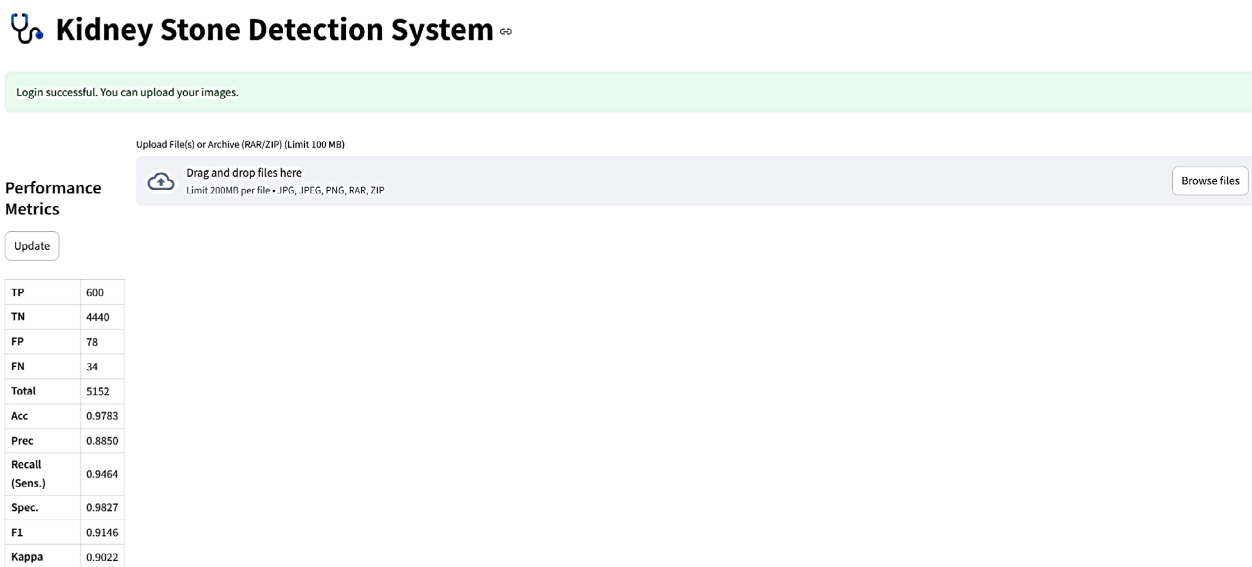


(a)



(b)

Fig. 4 Web-based interface of the pilot Kidney Stone Detection System. **a** Access layer interface of the pilot Kidney Stone Detection System showing the radiologist login screen. **b** Data and presentation layers of the Kidney Stone Detection System showing the file upload interface and real-time performance metrics

kernel was not consistently available across all cases and is therefore not reported.

All preprocessing, training, and internal validation procedures were completed prior to the pilot deployment described in this manuscript and have been reported separately. For the present work, a prospective pilot study was conducted over a six-month period, during which a total of 5,152 anonymized non-contrast CT slices from patients presenting with suspected renal colic were uploaded and evaluated through the web-based interface. Each participating radiologist independently annotated a distinct subset of cases, ensuring that no patient study was reviewed by more than one radiologist during the pilot phase. Only studies with complete abdominal coverage and adequate slice quality were included, while slices with severe artifacts or incomplete anatomical representation were excluded. All CT slices were de-identified prior to uploading them to the platform in accordance with institutional privacy policies.

Three board-certified radiologists participated in the pilot phase, each contributing independently to the case annotation process. Radiologists logged into the web-based interface and uploaded anonymized non-contrast CT slices of individual patients for evaluation, as illustrated in Fig. 5. Each CT study was annotated by a single board-certified radiologist, and no overlapping reads or adjudication process was implemented during the pilot phase. Accordingly, radiologist annotations were treated as a single-reader reference standard for model evaluation. Each uploaded study could contain numerous slices without visible renal structures; therefore, such irrelevant slices needed to be excluded from the annotation workflow. To automate this process, a pre-screening module developed in previous work [27]—based on an ANN classifier utilizing features extracted from ResNet18 and ViT encoders—was integrated into the interface to automatically filter out non-renal slices, substantially reducing manual workload and improving overall efficiency during the pilot study. As a result, radiologists could focus only on slices containing kidney regions, while a small number of residual non-renal slices that were not automatically detected could still be manually removed using the delete function (x icon) within the interface. After completing the annotation for a slice, the system revealed the AI-generated prediction for the corresponding slice, enabling immediate comparison and feedback, as shown in Fig. 5. All reported performance metrics therefore reflect agreement with a single reader rather than definitive diagnostic accuracy. All radiologist annotations, AI outputs, timestamps, and interaction logs were automatically stored for subsequent analysis.

Evaluation metrics and statistical analysis

During the pilot phase, the agreement between the deployed AI system and a single radiologist was prospectively evaluated by comparing AI predictions with radiologist-provided binary labels on a case level. Confusion matrices were constructed based on true positive (TP), true negative (TN), false positive (FP), and false negative (FN) outcomes obtained during the clinical simulation period. Standard classification metrics were computed in accordance with the grouped definitions presented in Eqs. (1)–(7). These included accuracy and balanced accuracy (Eq. 1); sensitivity (or recall) and specificity (or TNR), together with their complementary error rates FNR and FPR (Eq. 2 and Eq. 3); positive and negative predictive values—PPV and NPV (Eq. 4); the positive slice prevalence measure (Eq. 5); the F_1-score (Eq. 6); and Cohen's Kappa (κ), along with its associated observed and expected agreement components (Eq. 7). Together, these metrics quantify the agreement between the AI system and a single reader, reflecting agreement performance rather than inter-reader reliability or definitive diagnostic accuracy.

As no overlapping radiologist reads or adjudication process was implemented during the pilot phase, Cohen's κ should be interpreted as a measure of AI–single reader agreement. The 95% confidence intervals were estimated using the Wilson score method for binomial proportions. McNemar's test [28] was applied to assess whether observed differences in paired binary outcomes were statistically significant, with a significance level of $p < 0.05$. Comprehensive cross-validation and inferential analyses had already been performed during the model development stage. The present pilot evaluation complements these analyses by examining model behavior and clinician–AI agreement in a real-time interaction setting.

$$Accuracy\,(orAcc) = \frac{TP + TN}{TP + TN + FP + FN},\quad Balanced\ Acc = \frac{TPR + TNR}{2} \tag{1}$$

$$Recall\,(orTPR) = \frac{TP}{TP + FN},\ Specificity\,(orTNR) = \frac{TN}{TN + FP} \tag{2}$$

$$FNR = 1 - TPR,\ FPR = 1 - TNR \tag{3}$$

$$Precision\,(orPPV) = \frac{TP}{TP + FP},\ NPV = \frac{TN}{TN + FN} \tag{4}$$

$$Stone(+)Slice\ Prevelance = \frac{TP + FN}{TP + TN + FP + FN} \tag{5}$$

$$F_1 score = 2 \times \frac{TPR \times PPV}{TPR + PPV} \tag{6}$$

$$\kappa = \frac{P_o - P_e}{1 - P_e},\quad P_e = \frac{(TP + FP)(TP + FN) + (FN + TN)(FP + TN)}{(TP + TN + FP + FN)^2},\quad P_o = \frac{TP + TN}{TP + TN + FP + FN} \tag{7}$$

Experiments

Temporal stability of AI–single reader agreement during pilot deployment

Throughout the pilot study, radiologists continuously uploaded annotated CT cases to the web-based interface, enabling the system to compute and update AI–single reader agreement metrics as new data accumulated. Data entry and performance tracking were organized across six sequential evaluation points, representing periodic cumulative snapshots of the model's temporal behavior under real-world usage. The corresponding performance metrics computed at these six-time steps are presented in Table 2, where all

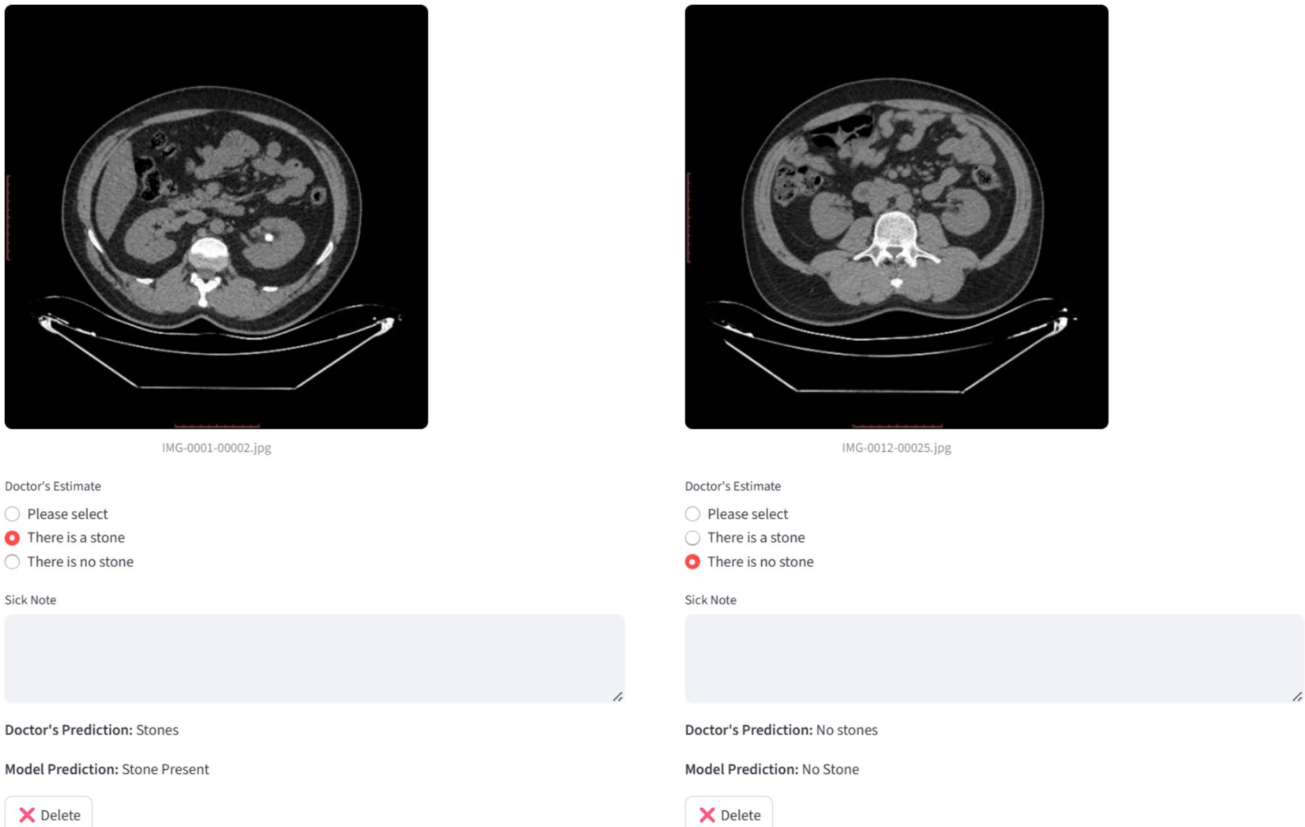


Fig. 5 Interactive review panel displaying doctor and AI model decisions on CT slices

confusion-matrix counts (i.e., the TP, TN, FP, and FN) are reported cumulatively up to each evaluation step.

As summarized in Table 2, the model exhibited a steady increase in agreement with a single reader as additional real-world cases were incorporated into the system. During the initial evaluation on 659 clinical slices, the model achieved an accuracy of 97.27%, precision of 78.38%, recall of 74.36%, specificity of 98.71%, F1-score of 76.32%, and a Cohen's Kappa of 0.7487, indicating moderate AI–single reader agreement. As the number of annotated slices increased, all agreement-related metrics demonstrated consistent upward trends, reflecting increasing stabilization of model behavior under real-world data diversity. This progression is further visualized in Fig. 6, where the temporal evolution of Cohen's Kappa illustrates the gradual stabilization of AI–single reader agreement across six successive evaluation stages. Notably, the most substantial changes occurred during the early pilot period, followed by gradual stabilization at a high level of agreement.

By the third evaluation point (3,556 slices), accuracy had risen to 97.81%, precision to 86.05%, recall to 90.80%, specificity to 98.51%, F1-score to 88.36%, and κ to 0.8715, indicating a marked enhancement in agreement between the AI system and the radiologist interpreting. Subsequent evaluations at the fourth (4,432 slices) and fifth (4,493 slices) stages revealed marginal yet consistent gains across all indicators, confirming that the model's agreement profile was approaching a stable regime. At the final evaluation step (5,152 slices), the model achieved 97.83% accuracy, 88.50% precision, 94.64% recall, 98.27% specificity, 91.46% F1-score, and a κ value of 0.9022, representing strong AI–single reader agreement rather than inter-reader reliability.

Importantly, Table 2 additionally reports balanced accuracy and stone-positive slice prevalence observed at each evaluation step to account for the strongly TN-dominated slice distribution encountered under real-world conditions. While overall accuracy remained high throughout the pilot, balanced accuracy increased from 86.54% at the initial evaluation step to 96.45% at the final step, indicating an improved sensitivity–specificity balance as case diversity increased. Over the same period, stone-positive slice prevalence rose from 5.91% to 12.31%, reflecting a gradual shift in case mix rather than changes in model parameters. The concurrent improvement in recall and balanced accuracy despite increasing prevalence suggests that the observed performance trends are not attributable to accuracy inflation driven by class imbalance but instead reflect stable and robust agreement behavior under evolving clinical data distributions.

This interpretation is further supported by the direct comparison presented in Table 3, which contrasts the online pilot phase with the offline hold-out evaluation. Despite a substantially lower stone-positive slice prevalence during the pilot (12.3%) compared with the hold-out set (60.1%), balanced accuracy remained comparable (96.45% vs. 97.13%), indicating that the model preserved a balanced detection performance under realistic, low-prevalence clinical conditions.

The observed difference in stone-positive slice prevalence across evaluation phases reflects differences in dataset construction and evaluation design rather than model bias or adaptation. During the development phase, the model was trained and evaluated using five-fold cross-validation on a total of 3,452 CT slices obtained from 235 cases, comprising 1,726 stone-positive slices from 196 patients and 1,726 stone-negative slices from 39 healthy subjects, resulting in an intentionally balanced slice-level distribution (50% prevalence). Importantly, all folds were constructed under strict patient-level separation to ensure that slices from the same individual did not appear in both training and validation sets. Model selection was performed based on cross-validation performance, and the selected model was subsequently evaluated on an independent hold-out set consisting of 732 slices (440 stone-positive slices from 45 patients and 292 stone-negative slices from 7 healthy subjects), which exhibited a higher stone-positive slice prevalence (60.1%).

The model finalized through this process was then deployed within the web-based platform for prospective pilot evaluation. In contrast to the offline evaluation settings, pilot data entry was performed on a study basis, whereby all slices belonging to a given patient were included in the evaluation stream. In routine abdominal CT examinations, only a limited number of slices per study typically contain visible calculi, while the majority represent stone-free anatomy. Consequently, stone-positive slices naturally constitute a small fraction of the total slice volume under real-world conditions, leading to a substantially lower observed prevalence during the pilot phase. This design choice was intentional, as it allowed the pilot evaluation to capture the inherent class imbalance encountered in clinical practice and to assess model robustness under realistic deployment conditions. Overall, these findings indicate that as the pilot progressed, the consistency between radiologist annotations and AI predictions improved, underscoring the importance of prospective workflow evaluations for assessing temporal stability of AI–single reader agreement in clinical settings.

As illustrated in Fig. 7, the trajectory of True Positive Rate (TPR) and False Positive Rate (FPR) across six pilot evaluation steps reflects the system's temporal stability under real-world deployment. As no retraining or parameter updates were performed during this period, changes in TPR

Table 2 Temporal evolution of classification performance across sequential evaluation steps

Time step	TP	TN	FP	FN	Total slice	Acc. (%)	PPV (%)	TPR (%)	Specificity (%)	Balanced acc. (%)	Stone + Slice prev. (%)	F1 (%)	κ
#1	29	612	8	10	659	97.27 ± **1.27**	78.38 ± **12.90**	74.36 ± **13.26**	98.71 ± **0.94**	86.53 ± **7.10**	5.92 ± **1.81**	76.32 ± **10.90**	0.75 ± **0.11**
#2	105	1647	29	21	1802	97.23 ± **0.76**	78.36 ± **6.92**	83.33 ± **6.49**	98.27 ± **0.63**	90.80 ± **3.56**	6.99 ± **1.18**	80.77 ± **5.25**	0.79 ± **0.06**
#3	296	3182	48	30	3556	97.81 ± **0.48**	86.05 ± **3.66**	90.80 ± **3.16**	98.51 ± **0.42**	94.66 ± **1.79**	9.17 ± **0.95**	88.36 ± **2.57**	0.87 ± **0.03**
#4	361	3977	62	32	4432	97.88 ± **0.43**	85.34 ± **3.37**	91.86 ± **2.72**	98.46 ± **0.38**	95.16 ± **1.55**	8.87 ± **0.84**	88.48 ± **2.29**	0.87 ± **0.03**
#5	422	3977	62	32	4493	97.91 ± **0.42**	87.19 ± **2.98**	92.95 ± **2.37**	98.46 ± **0.38**	95.71 ± **1.38**	10.10 ± **0.88**	89.98 ± **2.03**	0.89 ± **0.02**
#6	600	4440	78	34	5152	97.83 ± **0.40**	88.50 ± **2.40**	94.64 ± **1.77**	98.27 ± **0.38**	96.46 ± **1.08**	12.31 ± **0.90**	91.46 ± **1.56**	0.90 ± **0.02**

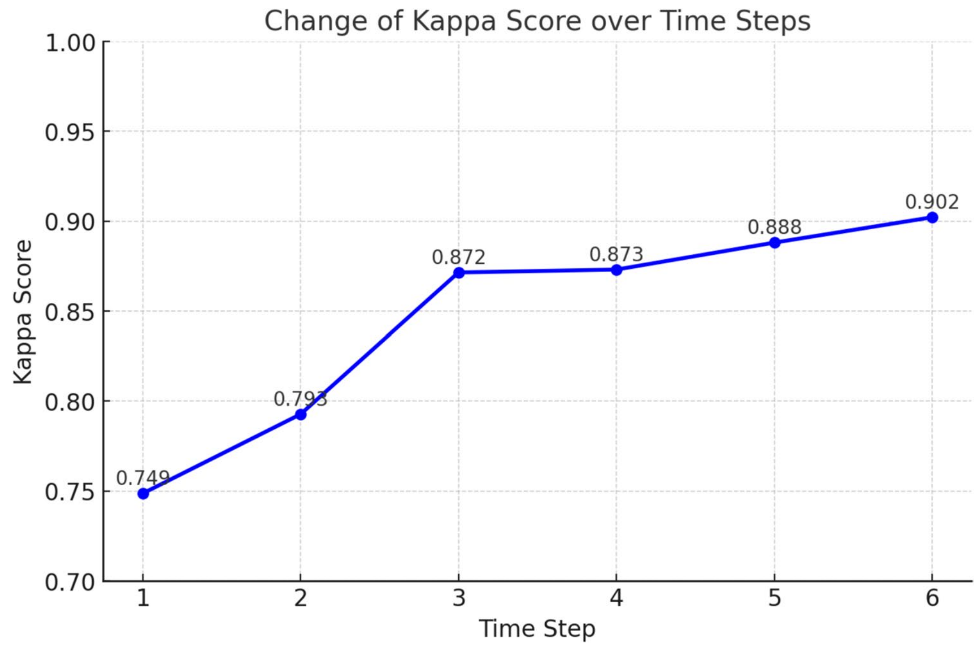


Fig. 6 Temporal change of Cohen's Kappa score during the pilot application process

Table 3 Comparison of stone-positive slice prevalence and balanced accuracy between the offline hold-out evaluation and the online pilot phase

Dataset	Stone-positive slices	Total slices	Stone + slice prev. (%)	Balanced acc (%)
Development phase (Offline)	440	732	60.1	97.13 ± 1.43
Pilot phase(Online)	634	5,152	12.3	96.46 ± 1.08

and FPR should be interpreted as reflecting variations in case mix and reader–AI interaction rather than model adaptation. Although TPR showed a gradual upward trend while FPR remained consistently low, this behavior indicates stable sensitivity–specificity characteristics of a fixed-threshold classifier over time, rather than performance optimization. In the early stages (#1–#2), lower TPR values likely reflect limited representation of case variability in the initial pilot cohort. As additional annotated CT slices were incorporated and case diversity increased, the observed TPR–FPR trajectory approached a stable operating region (TPR ≈ 0.95, FPR ≈ 0.015). Importantly, this visualization does not represent a conventional ROC analysis; instead, it illustrates the temporal behavior of a fixed classifier evaluated prospectively. The sustained balance between increasing TPR and stable FPR supports the robustness of AI–single reader agreement under continuous real-world evaluation.

Figure 8 illustrates how classification behavior differed between the controlled development environment (a) and the large-scale clinical pilot deployment (b). During model development, each abdominal CT study typically consisted of 40–60 axial slices, covering both kidneys and adjacent anatomical regions. In stone-positive patients, visible calculi were observed in only a few slices (usually 1–5), whereas the remaining majority represented stone-free parenchymal or background regions. This inherent imbalance led to a predominance of negative samples, which is reflected in the high number of true negative (TN) predictions in both matrices. In the model development phase (a), the model showed high agreement with the reference annotations, correctly matching 283 of 292 negative slices and 428 of 440 positive slices, yielding a balanced specificity (96.9%) and sensitivity (97.3%) under controlled conditions. However, this dataset was limited in size and relatively homogeneous, representing a laboratory evaluation setting. By contrast, the pilot evaluation phase (b) encompassed a substantially larger and more heterogeneous dataset, including approximately 5,000 test slices obtained from routine CT examinations with varying acquisition protocols and anatomical variability. Despite this increased diversity, the model preserved consistent agreement with single-reader annotations, producing 4,440 TN and 600 TP predictions, with a low number of false negatives (FN = 34) and a modest number of false positives (FP = 78).

The pronounced TN dominance observed in Fig. 8(b) underscores the model's consistent agreement with single-reader annotations for stone-free renal slices, a desirable characteristic in clinical workflows where excessive false-positive alerts could negatively affect efficiency. In contrast, the stable TP counts across both evaluation settings indicate

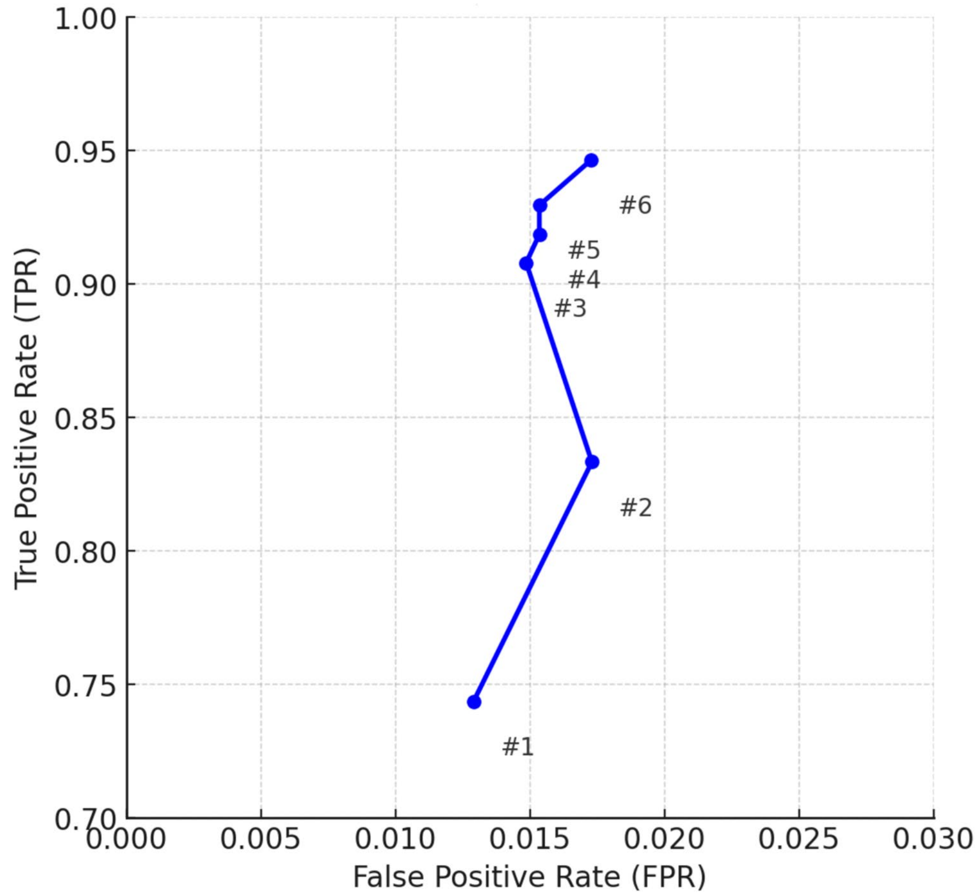


Fig. 7 Temporal stability of model performance in ROC space across six pilot evaluation steps

that the system maintained consistent agreement in identifying slices labeled as stone-positive by the reader, despite differences in dataset scale and clinical variability. Collectively, these findings suggest that the AI system preserved stable agreement behavior under real-world prospective testing conditions rather than reflecting changes in intrinsic model capability.

Figure 9 compares agreement-related performance metrics obtained during the offline model development stage with those observed during the pilot deployment. Based on the confusion matrices presented in Fig. 8, the model demonstrated high agreement with reference annotations during controlled offline evaluation (Accuracy = 0.97; κ = 0.94). Similarly, during the clinical pilot phase, agreement metrics remained high (Accuracy = 0.98; κ = 0.90), indicating temporal stability of AI–single reader agreement under routine-like conditions.

Compared with the offline stage, balanced accuracy showed a marginal decrease (0.97 → 0.96), while recall decreased slightly (0.97 → 0.95), reflecting differences in case mix and slice-level prevalence encountered during real-world deployment. Precision (PPV) and F1-score also decreased (PPV: 0.98 → 0.88; F1: 0.98 → 0.91), whereas negative predictive value increased (NPV: 0.96 → 0.99), indicating robust performance in excluding stone-free slices under low-prevalence clinical conditions. Importantly, specificity remained consistently high across both phases (0.97 → 0.98), further supporting the stability of agreement for negative cases. As no retraining or parameter updates were performed during the pilot phase, these metric shifts should be interpreted as reflecting variations in case composition and annotation distribution over time rather than model adaptation. Overall, the comparison confirms that the fixed model preserved stable agreement characteristics when transitioning from controlled development to prospective, workflow-aligned evaluation.

Study-level performance analysis on a patient-aware subset

While the prospective pilot evaluation in this study was primarily conducted at the slice level to reflect radiologists' routine slice review behavior and to maintain consistency with prior development and validation stages, study-level interpretation remains clinically more intuitive. Accordingly, the pilot phase was designed to prioritize slice-level annotation and performance tracking, enabling fine-grained monitoring of model behavior under real-world conditions.

Fig. 8 Comparison of confusion matrices between the model development phase (offline evaluation phase) **(a)** and the clinical pilot evaluation phase (or online evaluation phase) **(b)**

During pilot deployment, radiologists uploaded complete CT studies to the web-based platform and provided study-level case entries; however, annotation and performance logging were intentionally performed at the slice level. As a result, comprehensive study-level outcome derivation was not uniformly available across the entire pilot cohort, reflecting the original design focus of the pilot rather than a post hoc analytical limitation. Accordingly, all reported slice-level and derived study-level results from the pilot phase should be interpreted as agreement with a single reader, consistent with the single-reader annotation protocol employed throughout the deployment.

To partially address this design constraint and to improve clinical interpretability, an exploratory study-level aggregation analysis was conducted on a patient-aware subset for which slice–patient relationships and stone status were fully known. This subset consisted of 14 stone-positive patients

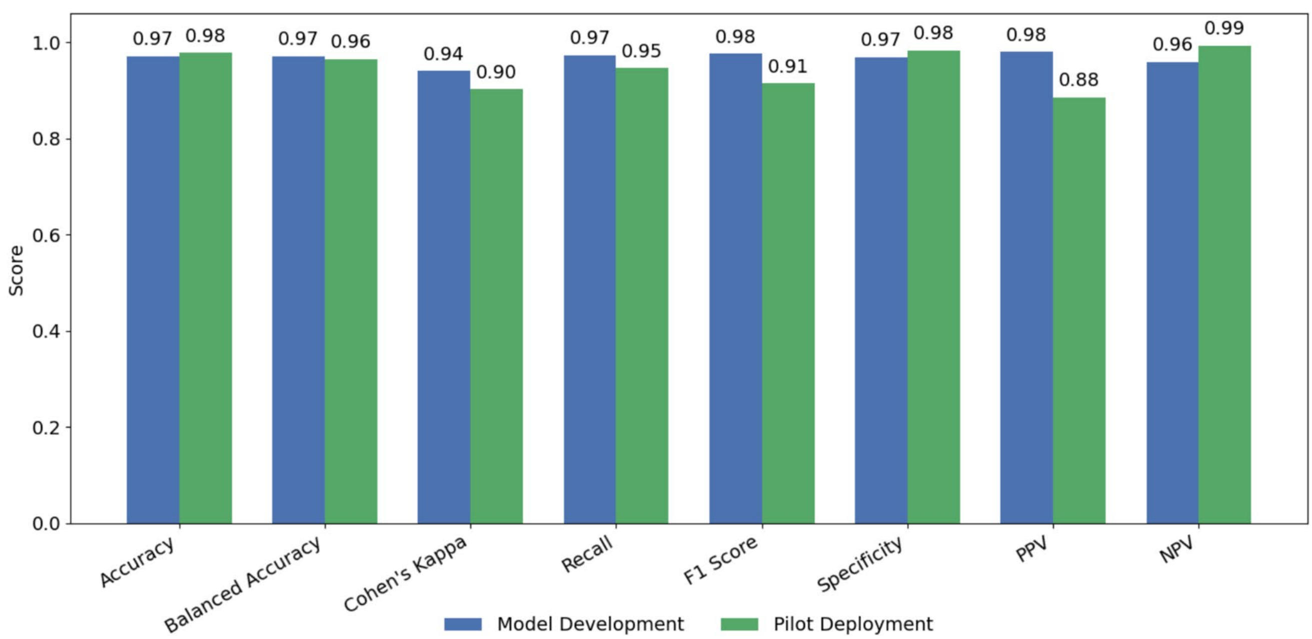


Fig. 9 Comparison of key performance metrics between the model development and pilot deployment phases

and 14 stone-negative (healthy) subjects, forming a balanced cohort selected from the available data.

For study-level inference, a conservative aggregation strategy was applied, whereby a CT study was classified as AI-positive if at least one slice within the study was predicted as stone-positive by the model. Under this rule, the presence of a single false-positive slice was sufficient for a study to be labeled as a false-positive at the study level. The resulting study-level confusion matrix, summarizing true-positive, true-negative, false-positive, and false-negative predictions, is presented in Table 4.

Using this aggregation approach, the model correctly identified all stone-positive patients (14/14), yielding no false-negative studies. Among the 14 stone-negative (healthy) studies, two were incorrectly classified as AI-positive (false-positive studies), as shown in Table 4. Detailed inspection of these false-positive studies revealed that the misclassification was driven by a very limited number of falsely flagged slices. Specifically, one study contained a single false-positive slice, while the other contained two false-positive slices, with all remaining slices in these studies correctly classified as negative and no false-negative slices observed.

Based on this confusion matrix, the derived study-level diagnostic performance metrics—including sensitivity, specificity, positive predictive value (PPV), negative predictive value (NPV), accuracy, balanced accuracy, false-positive rate (FPR), and false-negative rate (FNR)—are reported in Table 5. As summarized in Table 5, study-level aggregation resulted in a sensitivity of 100%, specificity of 85.7%, PPV of 87.5%, and NPV of 100%, indicating that slice-level predictions can be meaningfully translated into study-level decisions with a limited false-positive burden.

Nevertheless, given the limited sample size, the artificially balanced composition of the patient-aware subset, and the exploratory nature of this analysis, the reported study-level metrics should be interpreted with caution. In particular, PPV and NPV are prevalence-dependent and may not reflect real-world clinical distributions. Accordingly, this analysis is intended to provide an illustrative estimate of patient-level behavior rather than a definitive assessment of clinical diagnostic performance. A fully prospective study-level evaluation across a larger and unselected patient cohort will be required to obtain reliable prevalence-adjusted estimates.

Statistical significance of the pilot results

To quantitatively assess the diagnostic agreement between the AI system and radiologists during the final evaluation step (5,152 CT slices), a McNemar contingency matrix was constructed, as illustrated in Table 6. Because each radiologist evaluated a different, non-overlapping subset of cases,

Table 4 Study-level confusion matrix derived from slice-level predictions on a patient-aware subset

	AI Positive	AI Negative
Stone-positive studies	**14 (TP)**	**0 (FN)**
Stone-negative studies	**2 (FP)**	**12 (TN)**

Table 5 Exploratory study-level diagnostic performance metrics derived from slice-level predictions on a patient-aware subset

Metric	Score (%)	Observed case / Total case
Sensitivity (Recall)	100.0	14 / 14
Specificity	85.70	12 / 14
Positive predictive value (PPV)	87.50	14 / 16
Negative predictive value (NPV)	100.00	12 / 12
Accuracy	92.90	26 / 28
Balanced accuracy	92.85	13 / 14
False positive rate (FPR)	14.30	2 / 14
False negative rate (FNR)	0	0 / 14

inter-rater agreement among radiologists could not be computed. Consequently, all statistical comparisons reflect agreement between the AI system and individual radiologists rather than inter-reader diagnostic variability.

The matrix consisted of 600 true positives (both AI and radiologists correctly identified the presence of stones), 4,440 true negatives (both correctly identified stone-free slices), 78 false positives (AI predicted stones while radiologists did not), and 34 false negatives (radiologists detected stones that the AI missed). Based on these outcomes, the model achieved an accuracy of $97.83 \pm 0.40\%$, precision of $88.50 \pm 0.87\%$, recall of $94.64 \pm 0.62\%$, specificity of $98.27 \pm 0.36\%$, F_1-score of $91.46 \pm 0.76\%$, and κ of 0.90 ± 0.01, indicating strong agreement between the AI system and the interpreting radiologist. The narrow 95% confidence intervals across all metrics confirmed statistical stability under prospective evaluation conditions.

McNemar's test applied to the discordant pairs ($FP = 78$, $FN = 34$) yielded $\chi^2 \approx 16.5$ with $p < 0.001$, confirming that the imbalance between the two types of disagreements was statistically significant. Importantly, the predominance of false positives over false negatives reflects a conservative operating behavior that prioritizes sensitivity and minimizes missed stones, at the cost of a limited increase in review workload. Collectively, these findings demonstrate that the model preserved robust diagnostic consistency and reliable alignment with expert decisions throughout the final pilot phase.

Discussion

A growing number of studies have explored the use of artificial intelligence for automated kidney stone detection on CT slices [25, 29–36]. However, most of these efforts have concentrated primarily on algorithmic performance and retrospective dataset evaluations, with limited emphasis on workflow integration, radiologist interaction, or real-world deployment. Although many approaches report near-perfect accuracy on curated datasets, these results are typically derived under controlled and static conditions, lacking complexity, variability, and human feedback mechanisms characteristic of clinical practice. This limitation has also been acknowledged in broader medical informatics research, where scholars emphasize that the practical value of AI in healthcare arises not merely from algorithmic accuracy but from its integration, usability, and measurable impact within clinical systems [37, 38].

As summarized in Table 7, the few studies that have addressed deployment-oriented aspects remain confined to early stages of the Workflow Integration Level hierarchy—specifically the *Prototype Interface* level. Ramadhani and Salam (2024) integrated a YOLOv5-based object detection model into a Flask web application, achieving mean average precision values of 96.3% and 93.1% on mixed public and private CT datasets. While this demonstrated the technical feasibility of embedding AI models into web-based tools, the system was validated retrospectively and lacked any radiologist involvement [39]. Vardhan et al. (2025) proposed KidneyNet-V, a MobileNetV2 classifier embedded within a Streamlit web interface, which reached 99.7% accuracy on approximately 180 public CT slices [40]. Likewise, Abdalla et al. (2025) developed MyKidney, a Flask-based CNN system tested on over 5,000 public CT slices, reporting 98.8% accuracy under tenfold cross-validation [41]. Despite their strong numerical results, all three systems remained prototype demonstrations—operating offline, without clinician participation, and without longitudinal performance assessment.

In contrast, the present study progresses to the next maturity stage, classified as Simulated Workflow Integration. During the preceding 12-month development phase, a dual-stage CNN—first localizing kidney regions and subsequently classifying them—was developed and rigorously validated through fivefold cross-validation on 3,452 slices and tested on an independent set of 732 slices. The resulting model was then deployed within a Streamlit-based radiologist-facing web interface, enabling continuous case review, annotation, and performance monitoring throughout a six-month pilot period. During this phase, radiologists actively uploaded anonymized CT slices, annotated kidney stone presence, and compared their assessments with the model's predictions in real time. This setup simulated authentic diagnostic workflows within a controlled yet realistic environment, thereby enabling prospective evaluation of the AI system's diagnostic stability, temporal evolution, and concordance with human experts.

Temporal analyses conducted at six evaluation intervals revealed a consistent improvement in sensitivity, specificity, and Cohen's kappa, indicating progressive enhancement in model calibration and robustness as data diversity increased. Notably, this trend emerged *without retraining*, suggesting that the fixed model maintained stable agreement characteristics as data heterogeneity increased under operational conditions. The observed performance plateau beyond approximately 5,000 CT slices signified convergence, reflecting generalization across varying anatomical and imaging parameters.

From a diagnostic standpoint, the system consistently maintained a favorable balance between true-positive detection and false-positive suppression. Although each CT study typically comprised 40–60 axial slices, only a limited subset contained visible stones—resulting in a dataset with substantial class imbalance favoring negative slices. Despite this, the AI achieved high specificity and effectively identified stone-free slices, which is crucial for maintaining clinical workflow efficiency and minimizing alert fatigue. The model's strong true-negative discrimination thus underscores its reliability as a supportive diagnostic aid.

Comparison between the hold-out testing phase ($\kappa = 0.86$) and the pilot phase ($\kappa = 0.90$) demonstrated improved inter-rater agreement, confirming the stabilizing effect of radiologist-guided data exposure. This improvement highlights the necessity of prospective pilot evaluations prior to large-scale Clinical Deployment, as such environments capture behavioral variability that static retrospective analyses cannot. Although evaluation on an independent hold-out set provides evidence of robustness on unseen data, it does not constitute true external validation across institutions or acquisition protocols. Accordingly, generalizability beyond the source institution should be interpreted with caution and will require future multi-center validation.

Table 6 Contingency matrix between AI predictions and radiologist assessments at the final pilot evaluation step (used for McNemar's test analysis)

	AI: Positive	AI: Negative
Radiologist: Positive	600	34
Radiologist: Negative	78	4440

Table 7 Comparison of web-based AI systems for kidney stone detection and their workflow integration scope

Study	Deployed model	Platform/	Dataset origin	Test size	Workflow integration level	Radiologist involvement	Evaluation type	Reported accuracy (%)
Ramadhani & Salam (2024)	YOLOv5 (object detection)	Flask-based web app	Public [29] / Private CT	267/72	Prototype Interface	Not included	Offline retrospective testing	96.3/93.06
Vardhan et al. (2025)	MobileNetV2 (*KidneyNet-V*)	Streamlit web interface	Public CT [30]	~180	Prototype Interface	Not included	Offline retrospective testing	99.7%
Abdalla et al. (2025)	Custom CNN (*MyKidney*)	Flask-based web system	Public CT [41]	5319	Prototype Interface	Not included	Offline retrospective testing	98.8%
Present Study (2025)	Dual-stage CNN	Streamlit web interface	Private CT	5152	Simulated Workflow Integration	Active interaction via web-interface	Simulated prospective pilot (6 months)	97.8%

Workflow-oriented outcomes and practical utility

Beyond diagnostic performance metrics, several workflow-relevant observations emerged during the prospective pilot deployment. The pilot was not designed as a time-efficiency comparison between radiologists and the AI system; radiologists completed their slice-level assessments before AI outputs were revealed to ensure independent judgment and avoid anchoring bias. Accordingly, explicit time-to-label measurements were not collected, as the primary objective was prospective performance monitoring rather than speed benchmarking. Workflow utility was instead addressed through workload-oriented outcomes consistent with routine abdominal CT interpretation. An automated kidney-presence prescreening module filtered out slices without renal anatomy prior to stone assessment. Given that a typical non-contrast abdominal CT contains about 400–450 axial slices while renal anatomy is limited to only 80–100 of them, the prescreening step reduces the slice review volume by approximately 75–80%. This in turn yields an estimated ~64.1% reduction in annotation time, enabling radiologists to focus specifically on kidney-containing slices, as reported in previous study [27].

Beyond workload reduction, understanding the nature of errors introduced or mitigated by the proposed workflow is critical for assessing its clinical safety and interpretability. Accordingly, a qualitative error analysis was conducted to contextualize the practical impact of these outcomes. As summarized in Table 8 false-negative cases were predominantly associated with very small stones visible in only a single axial slice, as well as cases exhibiting limited stone–parenchyma contrast within the localized kidney region. These failure modes are primarily constrained by imaging resolution and acquisition-related factors and are therefore not directly addressed by region-of-interest localization.

In contrast, false-positive predictions were mainly related to extra-renal high-attenuation structures, including vascular calcifications, phleboliths, and bowel-related hyperdensities, which were occasionally emphasized by the regression-based localization stage when located adjacent to the kidney. Importantly, as reflected in Table 8, these error types are partially or likely mitigated by the proposed workflow through suppression of non-renal regions, and when present, predominantly resulted in additional slice review rather than missed diagnoses. This behavior reflects a conservative operating profile aligned with clinical triage priorities.

Limitations

Although the pilot environment replicated clinical interaction dynamics, it operated as a web-based simulation rather than a direct integration with hospital PACS (Picture Archiving and Communication System) or RIS (Radiology Information System) infrastructure. Radiologists participated remotely, contributing and reviewing anonymized CT cases via secure logins. This configuration preserved realistic evaluation and feedback mechanisms while preventing interference with routine clinical operations. Future work will target direct PACS integration, enabling seamless data exchange, automated case retrieval, and full regulatory validation under real-world operational settings. Remaining limitations include the single-center design, the limited number of radiologists participating, and the binary (stone/no-stone) classification scheme. Future extensions will incorporate multi-institutional data, 3D stone localization, and quantitative stone metrics (e.g., volume, attenuation) to yield richer clinical insights. Moreover, long-term deployment studies

Table 8 Representative error patterns and expected impact of the proposed workflow

Error type	Category	Typical scenario	Workflow impact	Expected mitigation by proposed model
FP	Vascular calcification	Extra-renal vascular calcifications adjacent to the kidney	Additional slice review	Likely reduced via suppression of non-renal regions by regression-based localization
FP	Phlebolith	Pelvic calcifications resembling distal ureteral stones	Additional slice review	Partially mitigated, depending on anatomical proximity to renal region
FP	Bowel content	Hyperdense bowel contents	Additional slice review	Likely reduced by exclusion of extra-renal abdominal regions
FN	Small stones	Very small stones visible in a single axial slice	Potential delayed detection	Not directly addressed, limited by imaging resolution
FN	Beam hardening	Local streak artifacts	Reduced stone conspicuity	Partially mitigated, but remains challenging under severe artifacts
FN	Noise	Low signal-to-noise ratio	Reduced confidence	Partially mitigated through ROI emphasis and dual-encoder feature fusion

will be necessary to monitor model drift and recalibration during extended operational use.

Overall, this study demonstrates that embedding AI systems within workflow-aligned web interfaces enables authentic prospective evaluation and facilitates meaningful human–AI interaction analysis. By extending beyond prototype-level demonstrations toward simulated workflow integration, the proposed framework provides a reproducible and systematic approach that bridges retrospective model development with real-world clinical implementation, thereby supporting the translational progression of AI-assisted diagnostic systems.

Conclusion

This study presents a structured and clinically oriented framework for advancing an AI-based kidney stone detection system from controlled retrospective development toward workflow-aligned prospective evaluation. Rather than aiming for full clinical integration—which requires direct PACS/RIS connectivity, regulatory certification, and hospital-embedded deployment—the present work focuses on an essential precursor stage: a simulated clinical workflow pilot conducted through a secure, browser-based platform.

By embedding a previously validated dual-stage CNN into a web interface designed for radiologist interaction, the study demonstrates that AI systems can be evaluated prospectively under realistic operating conditions without disrupting routine hospital processes. Throughout the six-month pilot period, three radiologists independently uploaded and annotated a total of 5,152 anonymized CT slices, allowing continuous assessment of model behavior, human–AI concordance, and temporal diagnostic stability.

The findings indicate that the model maintained strong and stable performance under real-world data variability, achieving 97.83% accuracy, 94.64% sensitivity, and a Cohen's κ of 0.90 by the end of the pilot. Importantly, diagnostic agreement exhibited increasing stabilization during the early pilot stages and approached a stable regime as case diversity increased. This evolution highlights the value of prospective workflow-aligned evaluation for identifying model strengths, calibration trends, and potential points of failure that retrospective analyses alone cannot capture.

While the pilot environment emulated clinical decision-making steps, it intentionally operated outside the PACS ecosystem and did not involve formal report generation. This design ensured safety, preserved radiologist independence, and enabled transparent performance monitoring, positioning the work at the Simulated Workflow Integration level identified in Table 1. As such, the study fills a critical gap between prototype-level demonstrations and full clinical deployment, providing empirical evidence that AI-assisted kidney stone detection can perform reliably when exposed to large-scale, heterogeneous, prospectively acquired imaging data.

Future efforts should focus on multi-center validation, direct PACS/RIS integration, regulatory compliance processes, and the extension of the system from binary slice-level classification to full 3D stone localization and quantification. Longitudinal monitoring during extended deployment will also be necessary to assess model drift and support continuous learning. Collectively, this work establishes a reproducible pathway for transitioning AI models toward eventual clinical adoption, demonstrating that workflow-oriented prospective pilots are a crucial intermediary step in the translation of radiological AI systems.

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Author contributions Coşku Öksüz: Conceptualization, Methodology, Software, Data curation, Formal analysis, Investigation, Visualization, Validation, Resources, Writing – original draft preparation, Writing—review & editing, Project administration, Funding acquisition.

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Data availability Due to patient privacy restrictions and data protection regulations, the dataset cannot be publicly shared. De-identified data may be made available upon reasonable request to the author, subject to institutional approval and compliance with relevant privacy and data governance guidelines.

Declarations

Conflict of interest The authors declare no competing interests.

Ethical approval Ethical approval for this study was obtained from the Clinical Research Ethics Committee of Kastamonu University (Approval No: 2023-KAEK-160, issued on 06/12/2023) in Turkey. All abdominal CT slices used in both the model development and pilot evaluation phases were fully anonymized at the source by removing all patient identifiers and protected health information prior to data transfer or analysis. All procedures complied with national regulations and institutional policies governing the ethical use of medical data in research.

Consent to participate Not applicable.

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